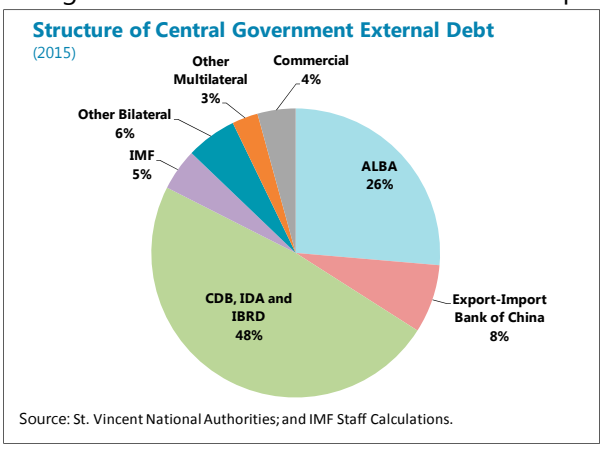
# Annex V. Debt Sustainability Analysis[1]

*In the absence of remedial measures, the risk of public debt distress has risen to high, even when abstracting from natural disasters and their additional fiscal burden, and other downside risks, which are not included in the baseline scenario. To better ensure a downward public debt path even in case of natural disasters, it is recommended to proactively adopt measures for fiscal consolidation and actively build a fiscal buffer to mitigate unexpected fiscal imbalances resulting from adverse shocks.*

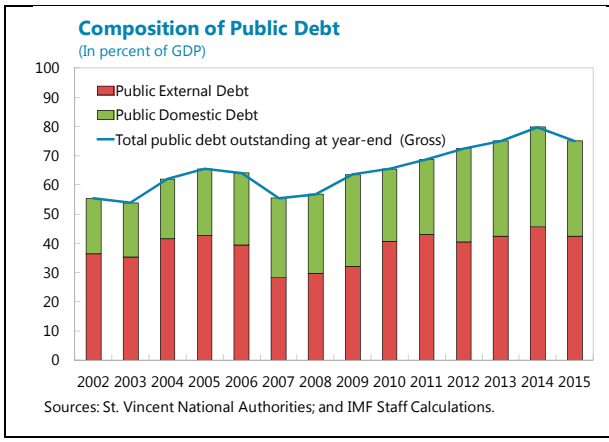
## A. Background

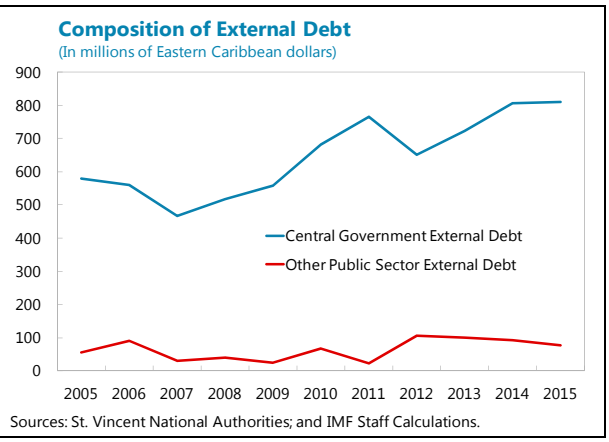
**St. Vincent's risk of public debt distress has increased to high under the baseline scenario where the authorities adopt no further fiscal measures.** Between 2008 and 2015, total public debt has risen from 57 to 73.4 percent of GDP, including to fund the construction of the new airport and the reconstruction in the aftermath of the end-2013 floods. The PetroCaribe agreement has represented an important financing source for the government. At end-January 2016, the debt of PetroCaribe St. Vincent to Venezuela stood at EC$183 million (8.7 percent of GDP), of which only EC$71 million is recorded as government debt. The other EC$112 million (5.3 percent of GDP) has not been entered in the government's public debt statistics, but is included in IMF projections. After peaking in 2018, total public debt is projected to decline—provided that the authorities do not take any additional fiscal measures to the ones already in place—to 71.3 percent of GDP by 2030, which is the year when ECCU countries have committed to achieve a goal of 60 percent in the public debt to GDP ratio.



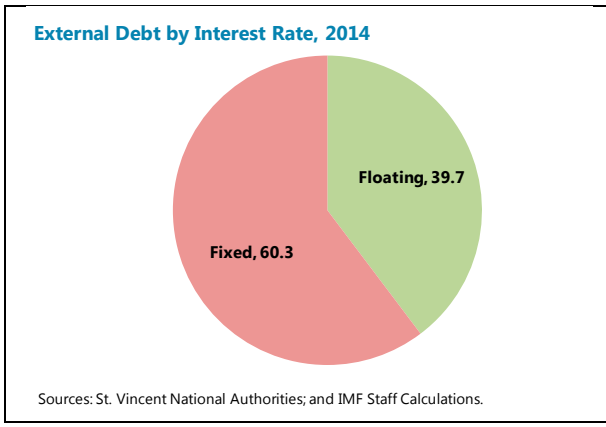
Source: St. Vincent National Authorities; and IMF Staff Calculations.

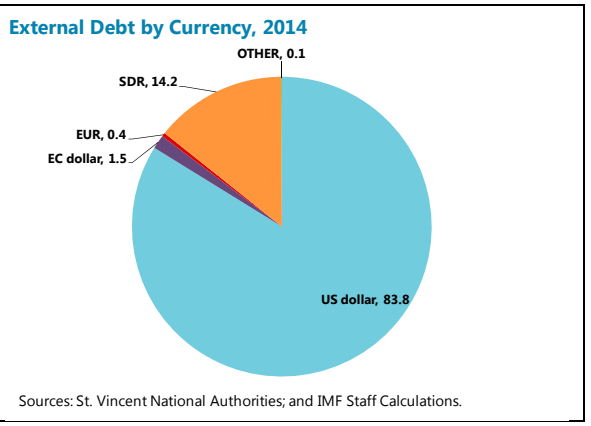
[1] St. Vincent and the Grenadines, classified as an upper-middle income country, has an IDA-blend status from the World Bank. Staff has been using the LIC country DSA.




**Most public debt is financed by external concessional financing, with the share of the central government rising until 2015.** Public debt reached 73.4 percent of GDP in 2015, of which 43.2 percent of GDP as external debt, and 30.1 percent of GDP as domestic debt. Central government obligations have been rising until 2015, and amounting to 90 percent the public external debt. In contrast, other public sector external debt has been declining in recent years. Most external debt is in multilateral and bilateral debt—the Caribbean Development Bank and Venezuela's ALBA Bank being the main creditors—with commercial external debt representing just 4 percent of the obligations.

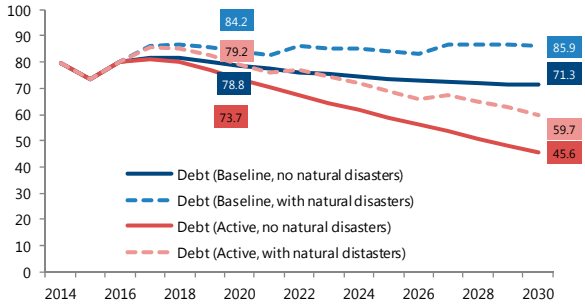



**While the latest available data shows that most of the debt is protected from exchange rate risk, interest rate risk is more of a concern.** In 2014, 40 percent of the external debt had been contracted at variable rates, which would be susceptible to a re-adjustment after an increase in US interest rates.

## B. Baseline Policy Scenario

**Under this scenario, the authorities adhere to their policies adopted to date without taking any additional measure.** The medium-term primary fiscal surplus of the public sector is projected to be 1.5 percent of GDP. The output gap is expected to close by 2021, with real GDP growth approaching 2.9 percent in the long run. Inflation, which has been historically low in St. Vincent, is expected to remain low in the medium term. The resulting public debt trajectory would decline to 71.3 percent of GDP, well above from 60 percent of GDP by 2030, the common ECCU target.




Source: Fund staff estimates.
[1] Assumes that natural disasters occur at 5-year intervals, lowering GDP growth in the first year and causing a 3-year primary balance deterioration associated to fiscal costs.

## C. Introducing Natural Disasters

**The occurrence of a natural disaster is modeled as an additional stress test.** The average natural disaster in St. Vincent and the Grenadines causes 5.5 percent of GDP in damages. A disaster of this magnitude would typically result in a drop in GDP growth of about 2 percentage points the year of the disaster, which accelerates by 0.5 percentage points the year after the disaster (boosted by reconstruction). Based on the past distribution of damages between the public and the private sectors, it is assumed that the government would cover 4 percent of GDP of the damages, and this additional fiscal expenditure is assumed to be spread over 3 years (2 percent of GDP in the first year, 1.2 percent of GDP in the second year and 0.8 percent of GDP in the third year). Additionally, revenues are assumed to be reduced by the fall in economic activity. Our representation of natural disasters considers a sequence of recurrent negative shocks (every 5 years) starting in 2017 which coincides with the second year of additional fiscal adjustments.

**In the case of recurring natural disasters, the baseline policy scenario results in an increasing public debt trajectory** which ends at 86 percent of GDP in 2030. This growing debt also enhances its vulnerability to simultaneous shocks of another nature.

**The annual contribution towards an amount that would adequately cover the fiscal cost of natural disasters is calibrated to the historical record for natural disasters in St. Vincent and the Grenadines.** Based on these records, staff estimates[2] that St. Vincent and the Grenadines would need to improve its primary fiscal balance by an additional 0.9 percent of GDP each year to adequately buffer a natural disaster. This is based on EM-DAT data showing that, between 1970 and 2014, St. Vincent experienced 10 natural disasters (3 floods and 7 storms) which caused a total of

[2] For more details, see Acevedo, S., "Gone with the Wind: Estimating Hurricane and Climate Change Costs in the Caribbean" (forthcoming IMF Working Paper).

55.3 percent of GDP in damages. Hence, on average, the country lost 1.2 percent of GDP annually from 1970 to 2014. Furthermore, the damage and loss assessments from the most recent disasters (hurricane Tomas in 2010, April floods in 2011, and December floods in 2013) show that the fiscal burden associated to the damages represents 73 percent of the total.

## D. Active Policy Scenario

**This scenario considers proactive fiscal measures to reach the ECCU-wide public debt target by 2030, including with building buffers to mitigate the fiscal impact of potential natural disasters.** The key difference between the active and baseline policy scenarios is the budgetary savings of 1.7 percent of GDP of which 0.9 percent of GDP assigned each year to eventually address the fiscal cost of natural disasters. Under the active policy scenario, in the absence of natural disasters, public debt is projected to decline steadily to 60 percent of GDP by 2025. However, when a recurrent natural disaster occurs, the debt still follows a downward trajectory but declines more slowly, reaching 60 percent of GDP by 2030.

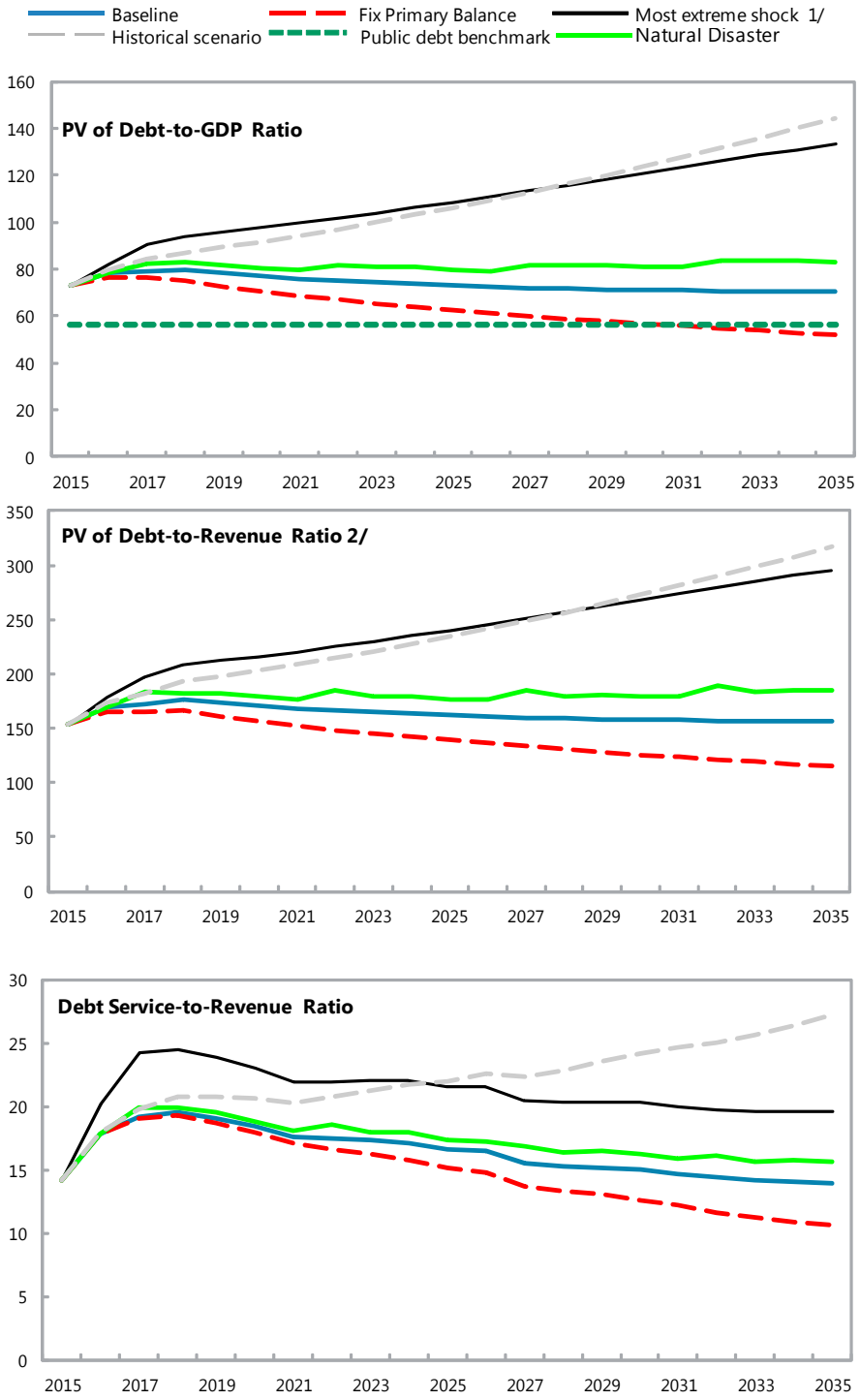
**Fiscal measures supporting the active policy scenario aim to restrain expenditure and enhance tax revenue.** These measures range from limiting wage bill increases, including by intensifying attrition, to tax policy changes such as raising the customs charge and curbing ad hoc exemptions. They also feature accelerating measures to improve revenue administration and public financial management. While revenue administration measures have strong potential to increase the yield of the VAT, income and profit taxes, and customs duties, these gains are not reflected in the fiscal projections. Similarly, although reforms in public financial management are envisaged to yield budgetary savings by improving the efficiency of public outlays, these savings are not reflected in the fiscal projections. The expenditure and tax policy measures are envisaged to contribute to a medium-term primary surplus of the public sector of 3.2 percent of GDP, 1.7 percent of GDP in budgetary savings added to the medium-term primary surplus of the public sector of 1.5 percent of GDP under the baseline policy scenario.[3]

## E. External Debt Sustainability Analysis

**The projected improvement in the current account balance over the medium-term is projected to place external debt on a downward trajectory**. While external debt represents 43.2 percent of GDP in 2015, staff projects that it would narrow to less than 16 percent of GDP by 2030. This is driven by improvements in the current account balance caused by imports declining at a faster pace than exports, and transfers mirroring the GDP growth rate. While external debt is expected to decline over time, it remains vulnerable to potential adverse shocks. Under the scenario with key variables held at historical levels, external debt would be on an upward trajectory, although this is not considered a likely outturn.

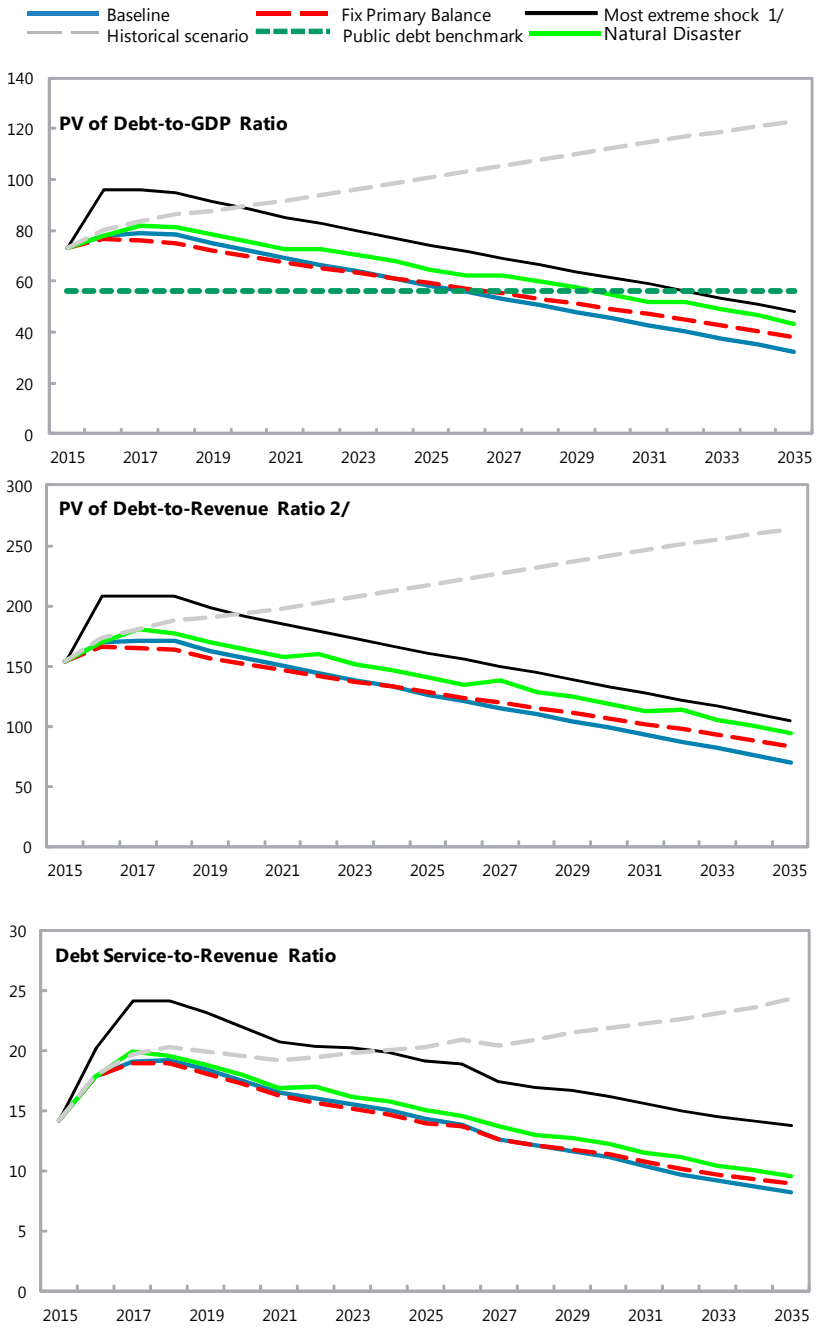
[3] This corresponds to a primary surplus of the central government of 1.6 percent of GDP under the current policies baseline scenario and a primary surplus of 3.3 percent of GDP under the active policy scenario.

**Figure A1. St. Vincent & the Grenadines: Indicators of Public Debt Under Baseline Alternative Scenarios, 2015-2035 1/**



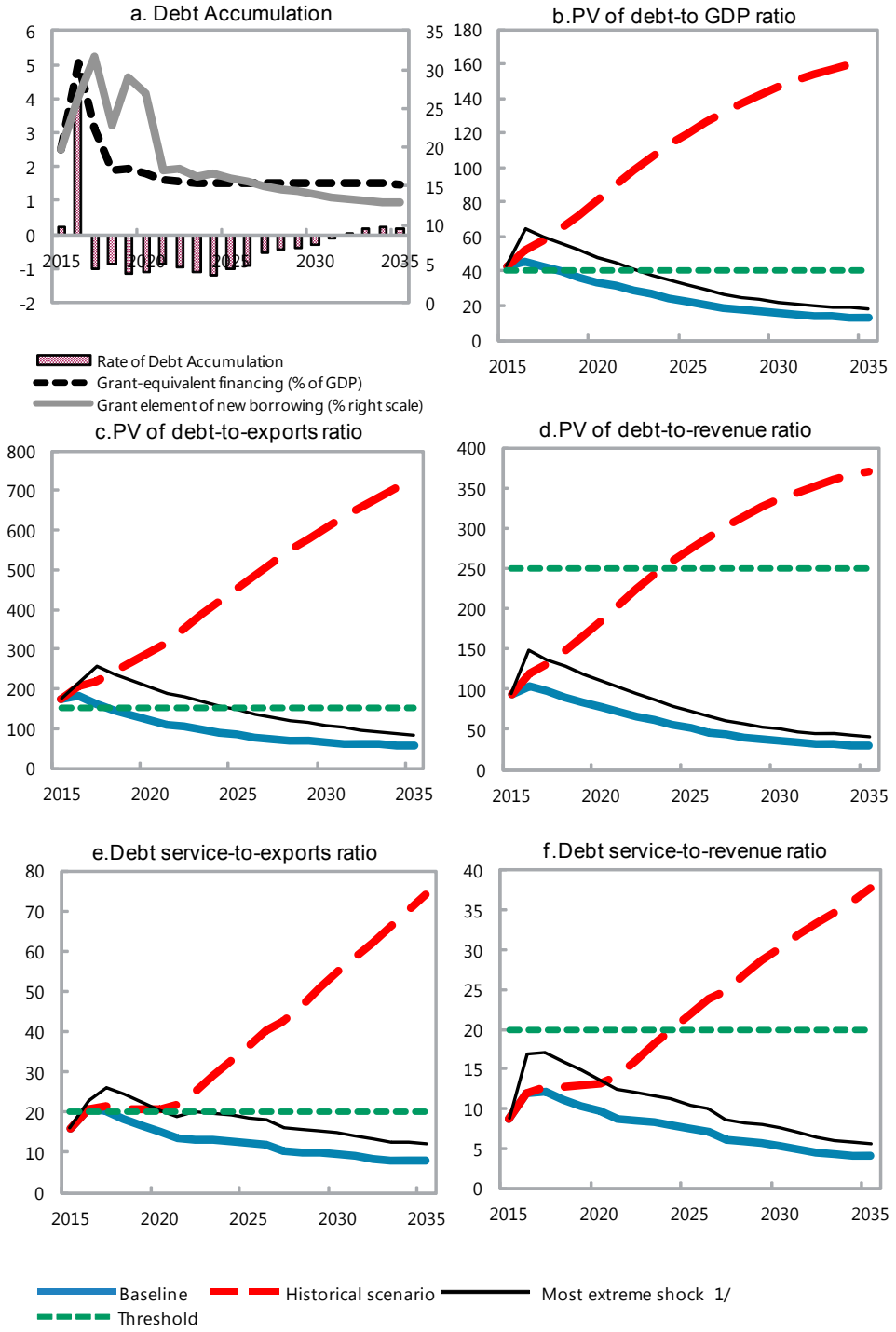

Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2025.
2/ Revenues are defined inclusive of grants.

**Figure A2. St. Vincent & the Grenadines: Indicators of Public Debt Under Active Alternative Scenarios, 2015-2035 1/**




Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2025.
2/ Revenues are defined inclusive of grants.

**Figure A3. St. Vincent & the Grenadines: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2015-2035 1/**




Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2025. In figure b. it corresponds to a One-time depreciation shock; in c. to a Combination shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

## Table A1. St. Vincent & the Grenadines: Public Sector Debt Sustainability Framework, Baseline Scenario, 2012-2035

(In percent of GDP, unless otherwise indicated)

| | Actual | | | | | Est. | Projections | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2012 | 2013 | 2014 | Average 5/ | Standard Deviation 5/ | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2015-20 Average | 2025 | 2035 | 2021-35 Average |
| **Public sector debt 1/** | 72.0 | 74.7 | 79.5 | | | 73.4 | 80.1 | 81.5 | 81.6 | 80.2 | 78.8 | | 73.6 | 70.6 | |
| *of which: foreign-currency denominated* | 40.5 | 42.3 | 45.3 | | | 43.2 | 47.3 | 44.5 | 41.6 | 38.5 | 35.6 | | 23.0 | 13.0 | |
| Change in public sector debt | 3.2 | 2.7 | 4.8 | | | -6.2 | 6.8 | 1.4 | 0.1 | -1.4 | -1.4 | | -0.8 | -0.1 | |
| Identified debt-creating flows | 3.4 | 0.5 | 1.3 | | | -2.6 | -0.1 | -0.6 | -0.6 | -1.3 | -1.2 | | -0.6 | -0.1 | |
| Primary deficit | 2.3 | 0.3 | -0.8 | 0.7 | 1.8 | -2.5 | -0.6 | -0.7 | -0.8 | -1.5 | -1.5 | -1.3 | -1.5 | -1.4 | -1.5 |
| Revenue and grants | 44.8 | 47.1 | 48.4 | | | 47.6 | 46.1 | 45.9 | 45.1 | 45.1 | 45.1 | | 45.0 | 45.0 | |
| *of which: grants* | 3.8 | 3.6 | 4.1 | | | 2.0 | 2.5 | 2.1 | 1.3 | 1.3 | 1.3 | | 1.3 | 1.3 | |
| Primary (noninterest) expenditure | 47.1 | 47.3 | 47.6 | | | 45.0 | 45.6 | 45.2 | 44.3 | 43.6 | 43.6 | | 43.2 | 43.6 | |
| Automatic debt dynamics | 1.2 | 0.3 | 2.1 | | | -0.1 | 0.5 | 0.1 | 0.2 | 0.2 | 0.3 | | 1.2 | 1.3 | |
| Contribution from interest rate/growth differential | 0.9 | 0.3 | 2.0 | | | 0.4 | 0.6 | 0.0 | -0.1 | 0.0 | 0.1 | | 1.1 | 1.3 | |
| *of which: contribution from average real interest rate* | 1.8 | 1.9 | 1.8 | | | 1.7 | 2.1 | 2.4 | 2.4 | 2.5 | 2.5 | | 3.2 | 3.3 | |
| *of which: contribution from real GDP growth* | -0.9 | -1.6 | 0.2 | | | -1.3 | -1.5 | -2.4 | -2.5 | -2.5 | -2.4 | | -2.1 | -2.0 | |
| Contribution from real exchange rate depreciation | 0.3 | 0.0 | 0.1 | | | -0.5 | -0.1 | 0.1 | 0.2 | 0.2 | 0.2 | | ... | ... | |
| Other identified debt-creating flows | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Privatization receipts (negative) | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Recognition of implicit or contingent liabilities | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Debt relief (HIPC and other) | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Other (specify, e.g. bank recapitalization) | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Residual, including asset changes | -0.2 | 2.1 | 3.6 | | | -3.6 | 6.8 | 2.0 | 0.7 | -0.1 | -0.3 | | -0.2 | 0.0 | |
| **Other Sustainability Indicators** | | | | | | | | | | | | | | | |
| **PV of public sector debt** | ... | ... | 78.5 | | | 73.0 | 78.0 | 79.3 | 79.5 | 78.3 | 77.0 | | 72.9 | 70.2 | |
| *of which: foreign-currency denominated* | ... | ... | 44.3 | | | 42.9 | 45.2 | 42.3 | 39.5 | 36.6 | 33.8 | | 22.2 | 12.7 | |
| *of which: external* | ... | ... | 44.3 | | | 42.9 | 45.2 | 42.3 | 39.5 | 36.6 | 33.8 | | 22.2 | 12.7 | |
| PV of contingent liabilities (not included in public sector debt) | ... | ... | ... | | | ... | ... | ... | ... | ... | ... | | ... | ... | |
| Gross financing need 2/ | 11.9 | 11.4 | 10.4 | | | 4.2 | 7.7 | 8.1 | 8.0 | 7.1 | 6.8 | | 5.7 | 4.8 | |
| PV of public sector debt-to-revenue and grants ratio (in percent) | ... | ... | 162.3 | | | 153.5 | 169.1 | 172.7 | 176.5 | 173.7 | 170.9 | | 161.9 | 156.1 | |
| PV of public sector debt-to-revenue ratio (in percent) | ... | ... | 177.4 | | | 160.3 | 178.9 | 181.1 | 181.7 | 178.8 | 176.0 | | 166.7 | 160.7 | |
| *of which: external 3/* | ... | ... | 100.1 | | | 94.2 | 103.7 | 96.6 | 90.3 | 83.6 | 77.3 | | 50.9 | 29.1 | |
| Debt service-to-revenue and grants ratio (in percent) 4/ | 15.6 | 15.5 | 15.4 | | | 14.2 | 17.8 | 19.2 | 19.6 | 19.1 | 18.4 | | 16.6 | 13.9 | |
| Debt service-to-revenue ratio (in percent) 4/ | 17.1 | 16.8 | 16.8 | | | 14.8 | 18.8 | 20.1 | 20.1 | 19.7 | 19.0 | | 17.1 | 14.4 | |
| Primary deficit that stabilizes the debt-to-GDP ratio | -1.0 | -2.4 | -5.6 | | | 3.6 | -7.3 | -2.1 | -0.8 | -0.1 | -0.1 | | -0.7 | -1.4 | |
| **Key macroeconomic and fiscal assumptions** | | | | | | | | | | | | | | | |
| Real GDP growth (in percent) | 1.3 | 2.3 | -0.2 | 1.1 | 2.6 | 1.6 | 2.2 | 3.1 | 3.1 | 3.1 | 3.1 | 2.7 | 2.9 | 2.9 | 2.9 |
| Average nominal interest rate on forex debt (in percent) | 2.9 | 3.0 | 2.7 | 3.8 | 0.9 | 2.3 | 3.2 | 3.1 | 3.1 | 3.1 | 3.2 | 3.0 | 4.5 | 3.8 | 3.5 |
| Average real interest rate on domestic debt (in percent) | 5.2 | 4.6 | 4.3 | 5.0 | 2.4 | 3.3 | 4.3 | 5.4 | 5.6 | 5.6 | 5.5 | 5.0 | 5.6 | 5.6 | 5.5 |
| Real exchange rate depreciation (in percent, + indicates depreciation | 0.7 | 0.0 | 0.1 | -0.3 | 2.2 | -1.0 | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| Inflation rate (GDP deflator, in percent) | 1.2 | 1.6 | 1.5 | 2.4 | 2.5 | 2.1 | 1.4 | 1.3 | 1.7 | 1.7 | 1.8 | 1.7 | 1.8 | 1.8 | 1.8 |
| Growth of real primary spending (deflated by GDP deflator, in percer | 0.6 | 2.8 | 0.3 | 0.4 | 0.9 | -3.8 | 3.4 | 2.3 | 1.0 | 1.5 | 3.1 | 1.2 | 2.2 | 2.9 | 2.9 |

Sources: Country authorities; and staff estimates and projections.

1/ Refers to non-financial public sector gross debt. From 2016, reflects additional debt contracted with PetroCaribe but not previously recorded (EC$ 112 million or 5.3 percent of GDP in 2016).

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

## Table A2. St. Vincent & the Grenadines: Public Sector Debt Sustainability Framework, Active Scenario, 2012-2035

(In percent of GDP, unless otherwise indicated)

| | Actual | | | | | Est. | Projections | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2012 | 2013 | 2014 | Average 5/ | Standard Deviation 5/ | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2015-20 Average | 2025 | 2035 | 2021-35 Average |
| **Public sector debt 1/** | 72.0 | 74.7 | 79.5 | | | 73.4 | 80.1 | 81.2 | 80.1 | 76.9 | 73.7 | | 58.9 | 32.4 | |
| *of which: foreign-currency denominated* | 40.5 | 42.3 | 45.3 | | | 43.2 | 47.3 | 44.6 | 41.6 | 38.5 | 35.6 | | 23.0 | 13.0 | |
| Change in public sector debt | 3.2 | 2.7 | 4.8 | | | -6.2 | 6.8 | 1.1 | -1.1 | -3.2 | -3.2 | | -2.8 | -2.7 | |
| Identified debt-creating flows | 3.4 | 0.8 | 1.3 | | | -2.6 | -0.1 | -1.0 | -1.7 | -3.1 | -2.9 | | -2.6 | -2.7 | |
| Primary deficit | 2.3 | 0.3 | -0.8 | 0.7 | 1.8 | -2.5 | -0.6 | -1.1 | -1.9 | -3.2 | -3.2 | -2.1 | -3.2 | -3.2 | -3.2 |
| Revenue and grants | 44.8 | 47.1 | 48.4 | | | 47.6 | 46.1 | 46.2 | 45.8 | 46.2 | 46.2 | | 46.2 | 46.1 | |
| *of which: grants* | 3.8 | 3.6 | 4.1 | | | 2.0 | 2.5 | 2.1 | 1.3 | 1.3 | 1.3 | | 1.3 | 1.3 | |
| Primary (noninterest) expenditure | 47.1 | 47.3 | 47.6 | | | 45.0 | 45.6 | 45.0 | 43.9 | 43.0 | 43.0 | | 42.6 | 43.0 | |
| Automatic debt dynamics | 1.2 | 0.6 | 2.1 | | | -0.1 | 0.5 | 0.2 | 0.1 | 0.1 | 0.3 | | 0.9 | 0.4 | |
| Contribution from interest rate/growth differential | 0.9 | 0.6 | 2.1 | | | 0.4 | 0.6 | 0.2 | 0.1 | 0.1 | 0.1 | | 0.7 | 0.4 | |
| *of which: contribution from average real interest rate* | 1.8 | 2.2 | 1.9 | | | 1.7 | 2.1 | 2.3 | 2.2 | 2.2 | 2.4 | | 2.5 | 1.4 | |
| *of which: contribution from real GDP growth* | -0.9 | -1.6 | 0.2 | | | -1.3 | -1.5 | -2.1 | -2.1 | -2.1 | -2.3 | | -1.7 | -1.0 | |
| Contribution from real exchange rate depreciation | 0.3 | 0.0 | 0.1 | | | -0.5 | -0.1 | 0.0 | 0.0 | 0.0 | 0.2 | | ... | ... | |
| Other identified debt-creating flows | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Privatization receipts (negative) | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Recognition of implicit or contingent liabilities | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Debt relief (HIPC and other) | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Other (specify, e.g. bank recapitalization) | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Residual, including asset changes | -0.2 | 1.9 | 3.5 | | | -3.6 | 6.8 | 2.0 | 0.7 | -0.1 | -0.3 | | -0.2 | 0.0 | |
| **Other Sustainability Indicators** | | | | | | | | | | | | | | | |
| **PV of public sector debt** | ... | ... | 78.5 | | | 73.0 | 78.0 | 78.9 | 78.1 | 75.0 | 72.0 | | 58.2 | 32.1 | |
| *of which: foreign-currency denominated* | ... | ... | 44.3 | | | 42.9 | 45.2 | 42.4 | 39.6 | 36.6 | 33.8 | | 22.3 | 12.7 | |
| *of which: external* | ... | ... | 44.3 | | | 42.9 | 45.2 | 42.4 | 39.6 | 36.6 | 33.8 | | 22.3 | 12.7 | |
| PV of contingent liabilities (not included in public sector debt) | ... | ... | ... | | | ... | ... | ... | ... | ... | ... | | ... | ... | |
| Gross financing need 2/ | 11.9 | 12.0 | 10.5 | | | 4.2 | 7.7 | 7.7 | 6.9 | 5.3 | 4.9 | | 3.1 | 0.6 | |
| PV of public sector debt-to-revenue and grants ratio (in percent) | ... | ... | 162.3 | | | 153.5 | 169.1 | 170.9 | 170.6 | 162.4 | 155.8 | | 126.1 | 69.6 | |
| PV of public sector debt-to-revenue ratio (in percent) | ... | ... | 177.4 | | | 160.3 | 178.9 | 179.2 | 175.6 | 167.1 | 160.4 | | 129.7 | 71.6 | |
| *of which: external 3/* | ... | ... | 100.1 | | | 94.2 | 103.7 | 96.2 | 89.0 | 81.6 | 75.4 | | 49.6 | 28.4 | |
| Debt service-to-revenue and grants ratio (in percent) 4/ | 15.6 | 16.7 | 15.5 | | | 14.2 | 17.8 | 19.1 | 19.2 | 18.4 | 17.5 | | 14.3 | 8.2 | |
| Debt service-to-revenue ratio (in percent) 4/ | 17.1 | 18.1 | 17.0 | | | 14.8 | 18.8 | 20.0 | 19.8 | 18.9 | 18.0 | | 14.7 | 8.4 | |
| Primary deficit that stabilizes the debt-to-GDP ratio | -1.0 | -2.4 | -5.6 | | | 3.6 | -7.3 | -2.2 | -0.8 | 0.0 | 0.0 | | -0.4 | -0.5 | |
| **Key macroeconomic and fiscal assumptions** | | | | | | | | | | | | | | | |
| Real GDP growth (in percent) | 1.3 | 2.3 | -0.2 | 1.1 | 2.6 | 1.6 | 2.2 | 2.7 | 2.7 | 2.7 | 3.1 | 2.5 | 2.9 | 2.9 | 2.9 |
| Average nominal interest rate on forex debt (in percent) | 2.9 | 3.7 | 2.8 | 3.9 | 0.8 | 2.3 | 3.2 | 3.1 | 3.1 | 3.1 | 3.2 | 3.0 | 4.5 | 3.8 | 3.5 |
| Average real interest rate on domestic debt (in percent) | 5.2 | 4.6 | 4.3 | 5.0 | 2.4 | 3.3 | 4.3 | 5.1 | 5.1 | 5.0 | 5.5 | 4.7 | 5.5 | 5.6 | 5.6 |
| Real exchange rate depreciation (in percent, + indicates depreciation | 0.7 | 0.0 | 0.1 | -0.3 | 2.2 | -1.0 | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| Inflation rate (GDP deflator, in percent) | 1.2 | 1.6 | 1.5 | 2.4 | 2.5 | 2.1 | 1.4 | 1.6 | 2.1 | 2.2 | 1.8 | 1.9 | 1.8 | 1.8 | 1.8 |
| Growth of real primary spending (deflated by GDP deflator, in percer | 0.6 | 2.8 | 0.3 | 0.4 | 0.9 | -3.8 | 3.4 | 1.4 | 0.1 | 0.5 | 3.1 | 0.8 | 2.1 | 2.9 | 2.9 |

Sources: Country authorities; and staff estimates and projections.

1/ Refers to non-financial public sector gross debt. From 2016, reflects additional debt contracted with PetroCaribe but not previously recorded (EC$ 112 million or 5.3 percent of GDP in 2016).

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

## Table A3. St. Vincent & the Grenadines: External Debt Sustainability Framework, Baseline Scenario, 2012-2035 1/

(In percent of GDP, unless otherwise indicated)

| | Actual | | | Historical 6/ Average | Standard 6/ Deviation | Est. | Projections | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2012 | 2013 | 2014 | | | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2015-2020 Average | 2025 | 2035 | 2021-2035 Average |
| **External debt (nominal) 1/** | **40.5** | **42.3** | **45.3** | | | **43.2** | **47.3** | **44.5** | **41.6** | **38.5** | **35.6** | | **23.0** | **13.0** | |
| *of which: public and publicly guaranteed (PPG)* | 40.5 | 42.3 | 45.3 | | | 43.2 | 47.3 | 44.5 | 41.6 | 38.5 | 35.6 | | 23.0 | 13.0 | |
| Change in external debt | -2.6 | 1.9 | 3.0 | | | -2.1 | 4.1 | -2.8 | -3.0 | -3.1 | -2.9 | | -2.0 | -0.4 | |
| Identified net debt-creating flows | 9.9 | 7.2 | 10.1 | | | 5.5 | 2.6 | 0.1 | -1.4 | -1.9 | -2.4 | | -2.3 | -1.8 | |
| **Non-interest current account deficit** | **26.4** | **29.8** | **28.4** | **26.4** | **4.9** | **23.8** | **20.0** | **18.7** | **16.2** | **14.9** | **14.0** | | **12.3** | **10.7** | 12.0 |
| Deficit in balance of goods and services | 30.4 | 32.8 | 31.4 | | | 26.0 | 22.5 | 20.6 | 17.9 | 16.2 | 15.0 | | 13.1 | 11.0 | |
| Exports | 27.5 | 25.2 | 25.1 | | | 24.6 | 25.0 | 26.1 | 26.9 | 27.4 | 27.9 | | 26.4 | 22.2 | |
| Imports | 57.9 | 58.1 | 56.5 | | | 50.6 | 47.5 | 46.7 | 44.8 | 43.6 | 42.9 | | 39.6 | 33.2 | |
| Net current transfers (negative = inflow) | -3.4 | -1.9 | -1.8 | -2.3 | 0.9 | -2.0 | -1.9 | -1.9 | -1.8 | -1.7 | -1.7 | | -1.5 | -1.3 | -1.4 |
| *of which: official* | -1.9 | -1.1 | -1.1 | | | -1.0 | -1.0 | -1.0 | -1.0 | -0.9 | -0.9 | | -0.8 | -0.7 | |
| Other current account flows (negative = net inflow) | -0.7 | -1.2 | -1.1 | | | -0.3 | -0.6 | 0.0 | 0.1 | 0.4 | 0.6 | | 0.7 | 1.0 | |
| **Net FDI (negative = inflow)** | **-16.6** | **-22.1** | **-18.9** | **-16.7** | **4.6** | **-18.6** | **-17.8** | **-18.5** | **-17.6** | **-16.8** | **-16.4** | | **-15.0** | **-12.6** | -14.3 |
| **Endogenous debt dynamics 2/** | **0.2** | **-0.4** | **0.6** | | | **0.3** | **0.5** | **0.0** | **0.0** | **0.0** | **0.0** | | **0.4** | **0.1** | |
| Contribution from nominal interest rate | 1.2 | 1.2 | 1.1 | | | 1.0 | 1.4 | 1.4 | 1.3 | 1.2 | 1.2 | | 1.1 | 0.5 | |
| Contribution from real GDP growth | -0.5 | -0.9 | 0.1 | | | -0.7 | -0.9 | -1.4 | -1.3 | -1.2 | -1.1 | | -0.7 | -0.4 | |
| Contribution from price and exchange rate changes | -0.5 | -0.7 | -0.6 | | | ... | ... | ... | ... | ... | ... | | ... | ... | |
| **Residual (3-4) 3/** | **-12.5** | **-5.4** | **-7.1** | | | **-7.6** | **1.5** | **-2.9** | **-1.6** | **-1.1** | **-0.5** | | **0.3** | **1.4** | |
| *of which: exceptional financing* | 0.0 | 0.0 | -2.0 | | | 0.0 | 0.2 | 0.3 | 0.3 | 0.4 | 0.3 | | ... | ... | |
| PV of external debt 4/ | ... | ... | 44.3 | | | 42.9 | 45.2 | 42.3 | 39.5 | 36.6 | 33.8 | | 22.2 | 12.7 | |
| In percent of exports | ... | ... | 176.8 | | | 174.5 | 180.9 | 161.9 | 146.9 | 133.4 | 121.1 | | 84.1 | 57.3 | |
| **PV of PPG external debt** | **...** | **...** | **44.3** | | | **42.9** | **45.2** | **42.3** | **39.5** | **36.6** | **33.8** | | **22.2** | **12.7** | |
| **In percent of exports** | **...** | **...** | **176.8** | | | **174.5** | **180.9** | **161.9** | **146.9** | **133.4** | **121.1** | | **84.1** | **57.3** | |
| **In percent of government revenues** | **...** | **...** | **100.1** | | | **94.2** | **103.7** | **96.6** | **90.3** | **83.6** | **77.3** | | **50.9** | **29.1** | |
| **Debt service-to-exports ratio (in percent)** | **15.2** | **16.8** | **17.7** | | | **16.1** | **20.7** | **20.2** | **18.2** | **16.6** | **15.1** | | **12.2** | **7.8** | |
| **PPG debt service-to-exports ratio (in percent)** | **15.2** | **16.8** | **17.7** | | | **16.1** | **20.7** | **20.2** | **18.2** | **16.6** | **15.1** | | **12.2** | **7.8** | |
| **PPG debt service-to-revenue ratio (in percent)** | **10.2** | **9.8** | **10.0** | | | **8.7** | **11.9** | **12.0** | **11.2** | **10.4** | **9.6** | | **7.4** | **4.0** | |
| Total gross financing need (Billions of U.S. dollars) | 0.1 | 0.1 | 0.1 | | | 0.1 | 0.1 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Non-interest current account deficit that stabilizes debt ratio | 29.0 | 27.9 | 25.4 | | | 25.9 | 15.9 | 21.5 | 19.2 | 17.9 | 16.9 | | 14.3 | 11.1 | |
| **Key macroeconomic assumptions** | | | | | | | | | | | | | | | |
| Real GDP growth (in percent) | 1.3 | 2.3 | -0.2 | 1.1 | 2.6 | 1.6 | 2.2 | 3.1 | 3.1 | 3.1 | 3.1 | 2.7 | 2.9 | 2.9 | 2.9 |
| GDP deflator in US dollar terms (change in percent) | 1.2 | 1.6 | 1.5 | 2.4 | 2.5 | 2.1 | 1.4 | 1.3 | 1.7 | 1.7 | 1.8 | 1.7 | 1.8 | 1.8 | 1.8 |
| Effective interest rate (percent) 5/ | 2.9 | 3.0 | 2.7 | 3.8 | 0.9 | 2.3 | 3.2 | 3.1 | 3.1 | 3.1 | 3.2 | 3.0 | 4.5 | 3.8 | 3.5 |
| Growth of exports of G&S (US dollar terms, in percent) | 4.3 | -4.6 | 0.5 | 0.0 | 5.2 | 1.8 | 5.4 | 9.2 | 8.0 | 6.9 | 6.8 | 6.3 | 2.9 | 2.9 | 3.2 |
| Growth of imports of G&S (US dollar terms, in percent) | 6.6 | 4.2 | -1.5 | 4.6 | 9.1 | -7.0 | -2.8 | 2.8 | 0.5 | 2.2 | 3.3 | -0.2 | 2.9 | 2.9 | 3.0 |
| Grant element of new public sector borrowing (in percent) | ... | ... | ... | ... | ... | 19.6 | 26.5 | 31.8 | 22.7 | 28.9 | 26.8 | 26.1 | 16.1 | 12.8 | 14.8 |
| Government revenues (excluding grants, in percent of GDP) | 41.0 | 43.5 | 44.2 | | | 45.6 | 43.6 | 43.8 | 43.8 | 43.8 | 43.8 | | 43.7 | 43.7 | 43.7 |
| Aid flows (in Billions of US dollars) 7/ | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| *of which: Grants* | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| *of which: Concessional loans* | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Grant-equivalent financing (in percent of GDP) 8/ | ... | ... | ... | | | 2.5 | 5.0 | 3.1 | 1.9 | 1.9 | 1.8 | | 1.5 | 1.5 | 1.5 |
| Grant-equivalent financing (in percent of external financing) 8/ | ... | ... | ... | | | 55.8 | 42.0 | 59.5 | 48.1 | 55.3 | 56.4 | | 58.1 | 53.7 | 55.1 |
| *Memorandum items:* | | | | | | | | | | | | | | | |
| Nominal GDP (Billions of US dollars) | 0.7 | 0.7 | 0.7 | | | 0.8 | 0.8 | 0.8 | 0.9 | 0.9 | 0.9 | | 1.2 | 1.9 | |
| Nominal dollar GDP growth | 2.5 | 4.0 | 1.3 | | | 3.7 | 3.6 | 4.5 | 4.8 | 4.9 | 4.9 | 4.4 | 4.7 | 4.7 | 4.8 |
| PV of PPG external debt (in Billions of US dollars) | | | 0.3 | | | 0.3 | 0.4 | 0.3 | 0.3 | 0.3 | 0.3 | | 0.3 | 0.2 | |
| (PVt-PVt-1)/GDPt-1 (in percent) | | | | | | 0.2 | 4.0 | -1.0 | -0.9 | -1.2 | -1.1 | 0.0 | -1.0 | 0.2 | -0.5 |
| Gross workers' remittances (Billions of US dollars) | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| PV of PPG external debt (in percent of GDP + remittances) | ... | ... | 42.9 | | | 41.6 | 43.9 | 41.1 | 38.5 | 35.6 | 33.0 | | 21.7 | 12.5 | |
| PV of PPG external debt (in percent of exports + remittances) | ... | ... | 156.3 | | | 154.5 | 161.1 | 145.5 | 132.9 | 121.4 | 110.9 | | 77.4 | 52.7 | |
| Debt service of PPG external debt (in percent of exports + remittance | ... | ... | 15.7 | | | 14.3 | 18.5 | 18.1 | 16.4 | 15.1 | 13.8 | | 11.3 | 7.2 | |

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

# Annex VI. Measures for Fiscal Consolidation

**The authorities correctly consider that reforming the revenue administration provides considerable scope to enhance tax revenue**, currently only adequate to cover current primary outlays (Figure 3)**.** Guided by CARTAC recommendations, the authorities intend to develop the capacity of the Customs and Excises Department (CED) to clear imports arriving at the new airport. They also aim to improve the physical infrastructure of the Inland Revenue Department (IRD) and reform its procedures. They seek to strengthen the monitoring of tax compliance by harmonizing tax identification numbers with identifiers from other registries, and improving communication between the CED and IRD, and the National Insurance Scheme. Moreover, enforcement capacities should be buttressed, and penalties should be increased and extended. Finally, the functionality of the IRD's appeals mechanism needs to be perfected. While these measures should raise the yield of income and sales taxes, including the productivity of the VAT (as measured by the ratio of VAT revenue to GDP divided by the VAT standard rate), currently low by regional standards (Figure 3), they are not reflected into the revenue projections.

**Efficiency gains are needed in public expenditure, through improvements in public financial management.** CARTAC is also supporting the authorities in this area. Tightening expenditure controls requires staffing the internal audit unit at the Ministry of Finance, upgrading the financial management information systems and setting up a cash management system unit at the Treasury, including to prevent the accumulation of new arrears. The predictability and control of transfers to public entities would benefit from tightening the oversight of public enterprises and strengthening their reporting requirements and sanctions regime, along with extending monitoring to the National Insurance Scheme (NIS) and the Kingstown Town Board (KTB). Developing a more proactive liquidity management strategy would improve predictability in funds availability. The Draft Procurement Bill, currently under revision following the World Bank's suggestions, aims to upgrade the competitiveness and efficiency of procurement procedures. The Public Sector Investment Plan (PSIP) is being improved by a more systematic analysis of projects and their impact on recurring costs. The authorities expect that these elements will be published in the macro fiscal documents underpinning the 2017 budget. While these measures should contribute to budgetary savings while enhancing the efficiency of public spending, they are not reflected into expenditure projections.

**Reforms in tax and expenditure policy, politically more difficult to implement, would provide tangible support to the fiscal consolidation.**

- **The authorities are considering curtailing discretionary tax and customs concessions**, with the latter reportedly amounting to nearly 1.8 percent of GDP at end- 2015. While this policy action would be beneficial, including because it would facilitate revenue administration procedures, there is also scope for deeper reforms. In particular, a thorough review of concessions granted through special legislation (mainly for large and upscale tourism projects and the airport's development) with a publication of their budgetary costs (foregone revenue of Customs concessions is currently estimated at 3.8 percent of GDP) would be very helpful.

- **An important pillar of any fiscal consolidation is to contain the wage bill, which is relatively high in St. Vincent and the Grenadines, even by regional standards** (Country Report No. 14/360, paragraph 15)**.** The experience of smaller states has confirmed that successful and sustained episodes of large fiscal consolidation have relied on curbing the wage bill as a key component of their strategy.[1] While this can be effected for a few years by a simple wage and hiring freeze, a more durable, if lengthy, solution is to reform the civil service. This would include retrenching certain positions and re-deploying staff towards new positions. While the authorities have previously attempted a first step towards civil service reform— instituting a performance management system—without success, this may be a more viable option once the private sector recovers and offers additional employment opportunities.

- **Improving the sustainability of the public pension scheme and the National Insurance Service would be a key element of the fiscal consolidation.** Expenditure on pensions and retirement benefits have grown from 2½ percent of GDP in 2011 to 3 percent of GDP in 2015 and is expected to keep rising, since the five pension schemes of public servants are all non-contributory and unfunded. This risks undermining the medium-term fiscal position, which can be addressed by implementing parametric reforms among the options suggested by the actuarial analysis published in 2015.

[1] Simard, D., Lonkeng, C, and G. Lindow (2015) "Small States and Large Fiscal Adjustments: Recent Experience", in Caribbean Corner, Issue No. 04, September 2015, International Monetary Fund.